

2018

TEXAS SEVERE WEATHEAR DATA VISUALIZATION PROJECT

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SOUTHERN NEW HAMPSHIRE UNIVERSITY | 04/15/2018

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BACKGROUND AND AUDIENCE ANALYSIS

Severe weather events have the potential to wreak havoc on communities, and they show no regard for what is lost in their wake. Preparation is essential to efficient response and recovery. The Governor's office, the regional emergency management teams, and the public each have a part to play in preparation and recovery from a severe storm. Storm data collection and analysis are essential to overall effectiveness. Without the collection of weather and climate data, government officials can't determine the budget for weather-related emergency relief, and risk not having the resources to support their state. Without adequate grant funding, emergency management teams are unable to deploy recovery resources to their districts, and risk public panic. Without public awareness, families risk being without life sustaining resources during and following a weather disaster. All of which can hinder economic growth within the state of Texas. It is essential that Texas as a whole is capable of quickly recovering from the destruction caused by severe weather events.

The National Centers for Environmental Information (NCEI) is an environmental data government authority that develops and maintains datasets for current and historical national and global weather events (NCEI, 2018). The authority also monitors anomalies in climate caused by the depletion of the Earth's natural resources (NCEI, 2018). The data is available to the public. The data is comprehensive, as its sources include atmospheric data, marine data, land data, and paleoclimate data (NCEI, 2018). The focus of this project is severe weather. Severe weather includes, thunderstorms, tornados, hail storms, blizzards, and tropical systems (NCEI, 2018). NCEI provides several datasets focused on severe weather activity, including the Storm Events Database, Severe

Weather Data Inventory (SWDI), International Best Track Archive for Climate Stewardship (IBTrACS), and Lightning Products and Services (NCEI, 2018). This project utilized the Storm Events Database to create three visualizations for three different audience types. The project focus is storm activity for the state of Texas from January 2015 to December 2016. The intended audience includes; state officials, emergency management personnel, and the public.

THE DATASET

The Storm Events Database contains data dated from January 1950 to November 2017 (NCEI, 2018). The database focuses on storms with the potential for physical damage and the occurrences of significant climate anomalies (NCEI, 2018). The sources for the storm data are the National Weather Service, the media, government agencies, and the public (NCEI, 2018). All storm data must be certified by the Warning Coordinator Meteorologist (WCM) or Meteorologist-in-Charge (MIC) to be included in the dataset (MacAloney, 2016). The data was prepared using an online software program approved by the Office of Chief Operating Officer (OCOO) with the National Weather Service (MacAloney, 2016).

Great focus was given to the location and time of an event to ensure data was accurately documented and the location recorded matched the county listed (MacAloney, 2016). Any discussion of “episode” describes the entire storm system, and any discussion of “events” describe the storm type (NCEI, 2018). For example, a storm producing thunderstorm winds and floods is one episode with two events. An event is considered significant if the threshold criteria was met and/or the event was considered newsworthy due to the magnitude of its impact on surroundings (MacAloney, 2016). Non-weather-related fatalities and injuries that occurred on roadways during the storm

were not included in the dataset (MacAloney, 2016). Damage refers to physical property (MacAloney, 2016).

Event Details. An algorithm was used to calculate the longitude and latitude for the location of the storm event if not input directly (NCEI, 2018). In general, location is the nearest hundredth mile to the city center (MacAloney, 2016). The flood and debris flow are referenced by two location points with additional details regarding the full path of flow included in the narrative (MacAloney, 2016). All times listed in the dataset are in local standard and are reflective of the time the event established advisory or warning level (MacAloney, 2016). Winter events are the exception, as the beginning time of the event was the time precipitation accumulation began and end time was the time precipitation ended (MacAloney, 2016). Tornadoes are counted by segments (NCEI, 2018). For example, a single tornado crossing the state or county line is counted as two segments, and a tornado that lifts away from the ground for less than 5 minutes and/or less than 2 ½ miles is counted as separate segments (NCEI, 2018). Events that occur in two or more consecutive months are listed separately for each month active (MacAloney, 2016). All other events are of a single count.

AUDIENCE – STATE/GOVERNMENT OFFICIALS

The target audience are high level officials responsible for developing Texas budgetary policy and are involved in the allocation of emergency relief. The audience are highly educated members of government agencies. This includes local, city, and state officials. The goal for the project is to demonstrate to this audience the regions within the state most likely to require emergency relief in response to a severe storm event. This will ensure adequate funds and resources are allocated to the appropriate regions. To demonstrate the need, a PowerPoint presentation was developed. The presentation is

highly influential in the budget designation and is ideal for this audience because it is a static document that focuses on the audience needs and offers a solution that is quickly interpreted by the viewers (Hearn, 2017). The complete budget bill is reviewed by legislature during a specific time of the year, therefore, the audience has limited time to review the presentation and decide the budget designation for emergency relief (Texas Impact, 2015). To ensure adequate time for review, the presentation is concise. It contains charts and other supports designed for quick decision-making. Some officials may not be skilled in the technical aspects of storm activity; therefore, the presentation maintains common terminology that is easily understood by all viewers. Cultural diversity, age, and language are not relevant.

AUDIENCE – EMERGENCY MANAGEMENT PERSONNEL

The targeted audience are members of the Texas Division of Emergency Management. The audience is highly educated, knowledgeable of storm-related terminology, and technically skilled. This audience focus is storm activity for the counties under their division (TDEM, 2018). This differs from the first audience, where the focus is the entire state of Texas. The goal for the project is to design interactive dashboards utilizing the Storm Events Database. The purpose for the dashboards is to provide this audience with means for assessing risk by district, determining the appropriate allocation of emergency equipment and resources, and developing the appropriate response to media inquiries. The dashboard includes various types of charts, including color concentration maps by event type and location, a bar chart demonstrating high and low averages by county, and additional focus on storm events requiring media response. A dashboard is appropriate because the audience can interact with the

dashboard to develop risk assessments, then take that information and develop regional response plans for the counties under their division (TDEM, 2018). Cultural diversity, age, and language are not relevant.

AUDIENCE – THE PUBLIC

The targeted audience are members of the public. This includes residents, visitors, the media, and any member of the public with the potential to be directly or indirectly affected by storm activity within the state of Texas. The goal of this project is to design a webpage offering safety tips and guidance for severe storm preparedness. The purpose of the webpage is to inform the public of the risks associated with floods. The webpage includes interesting charts with simple statistics for added perspective, as well as a video demonstrating risks in action. In contrast to the first two audiences, this audience is not responsible for developing state budgetary policy or regional emergency response plans. This audience focus is personal safety and preservation of their property. Cultural diversity, education, age, and language for this audience vary. Therefore, the use of simple universal graphics is ideal. A webpage is appropriate for this audience because it is easily accessible from anywhere by anyone with internet access. The webpage offers quick access to information providing support that is essential to public safety when affected by severe weather events.

DATA VISUALIZATION STRATEGY

POWERPOINT PRESENTATION

Elements and Formatting

- Balanced and minimalistic style to ensure focus on charts. White space balanced to content. Brief bulleted talking points

- Two panel slides – text on left and charts on right. Charts are sized and positioned for greater focus. Text box is smaller than chart box for visual impact
- Bar charts arranged with the greatest visual impact on left (highs to lows) to follow path of sight for quick interpretation by the audience
- Consistent muted color pallet (light blue/green, dark blue, light gray, white, and dark orange)
- All slides and elements arranged to move with the flow of the transcript
- Last slide reserved for names and titles of project leaders, phone number/email for questions and feedback
- Reference (reduced to ensure all fit on one slide)

INTERACTIVE DASHBOARDS

Elements and Formatting

Three colorful dashboards with multiple charts demonstrated by district and county. Bright multicolor for the 23 districts, font Arial (11pt for data, 16pt for title)

Dashboard one – event type by district/county to assist in assessing risk levels.

Drill down by district and county. Content and layout are as follows.

Legend colored by district. Includes total event count by district	
Map chart reduced visual to Texas, colored by district with point plots sized by event count for the district. Zoom in/out included	
Bar chart colored by district and sized by event count for the district. Legend colored by district	Pivot chart detailing counties per district. Same font, color lavender and purple when selected
Author & affiliation, date, contact information, link to feedback survey and dataset	

Dashboard two – recovery costs and frequency of event type by district to determine appropriate allocation of emergency equipment and resources. Drill down by district and county. Content and layout are as follows.

Legend colored by district and includes total recovery cost for the district
Bar chart detailing cost by district. Single gradated color by high to low cost per district. Arranged from high to low cost per district.
Multiple mini bar charts (23) detailing top event type by district. Each chart color by district. Same colors as dashboard one
Author & affiliation, date, contact information, link to feedback survey and dataset

Dashboard three – event effects by district/county to aid in developing the appropriate response to media. Drill down by district, effect type, and county. Content and layout are as follows.

Legend colored by district and included total counts for deaths, injuries, property damage, and crop damage by district			
Color/bubble scatter plot sized by death count by district.	Color/bubble scatter plot sized by injury count by district.	Color/bubble scatter plot sized by damaged property count by district.	Color/bubble scatter plot sized by damaged crops count by district.
Pivot chart detailing county by all four tragedy/effective count. Same font. Includes column with media event details for response (full detail pop-out when selected)			
Author & affiliation, date, contact information, link to feedback survey and dataset			

WEBPAGE

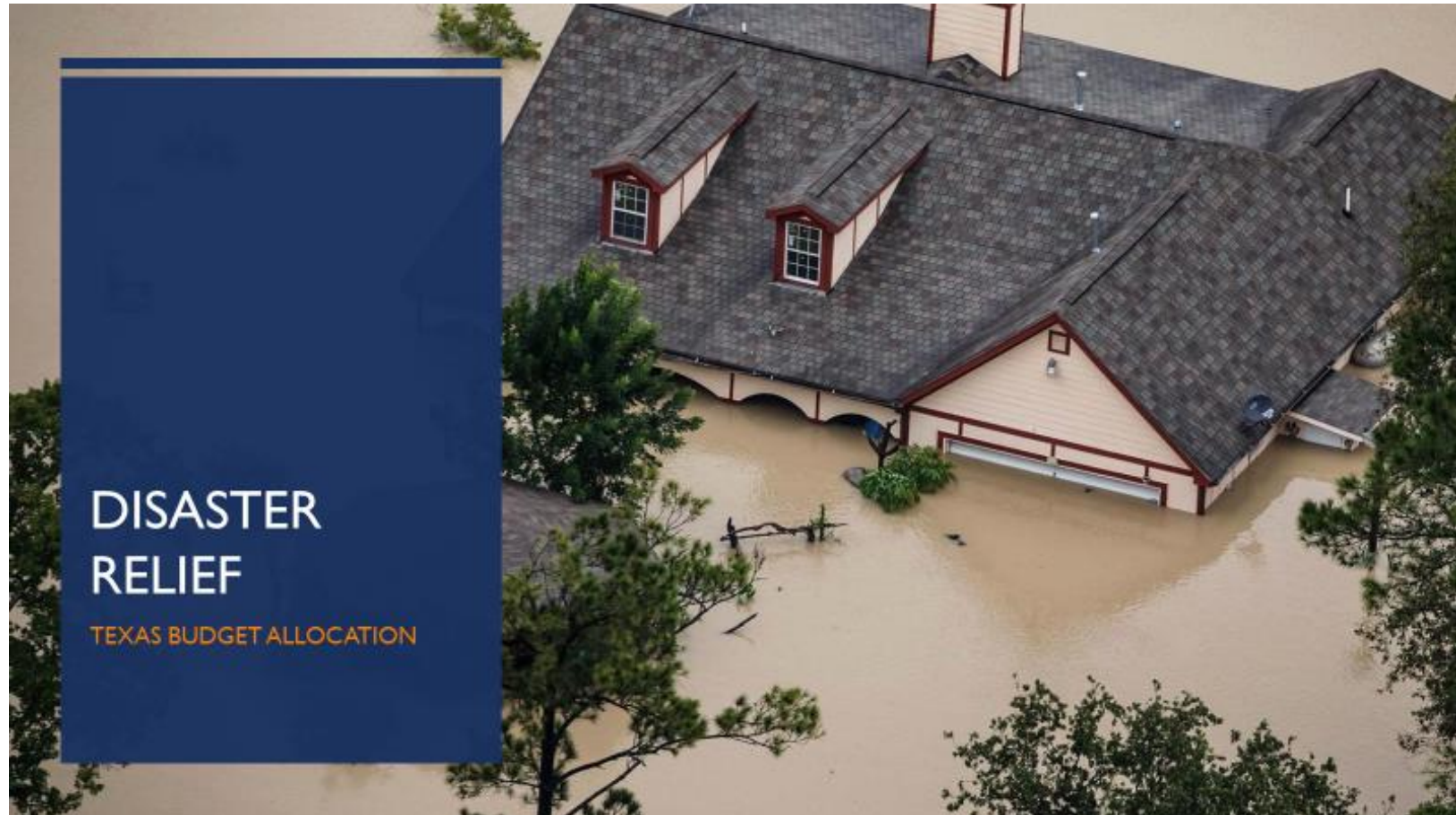
Elements and Formatting

- Overall style is informative
- Two sectioned title - Texas logo left-justified followed by title in large font (intense emphasis)
- Color pallet red, white, and blue to complement Texas logo – color is somewhat muted due to the high contrast colors, but bright enough to attract interests of all backgrounds

- You tube video - flood victim (centered)
- Embedded page links to page sections below (small, black font)
- All embedded citations are linked to reference at bottom of page
- Static chart demonstrating simple statistics regarding vehicle deaths by flood in Texas
- Multiple paragraphs detailing tips and guides for flood safety
- Author name, date posted, link to feedback survey
- To-top button
- Reference list

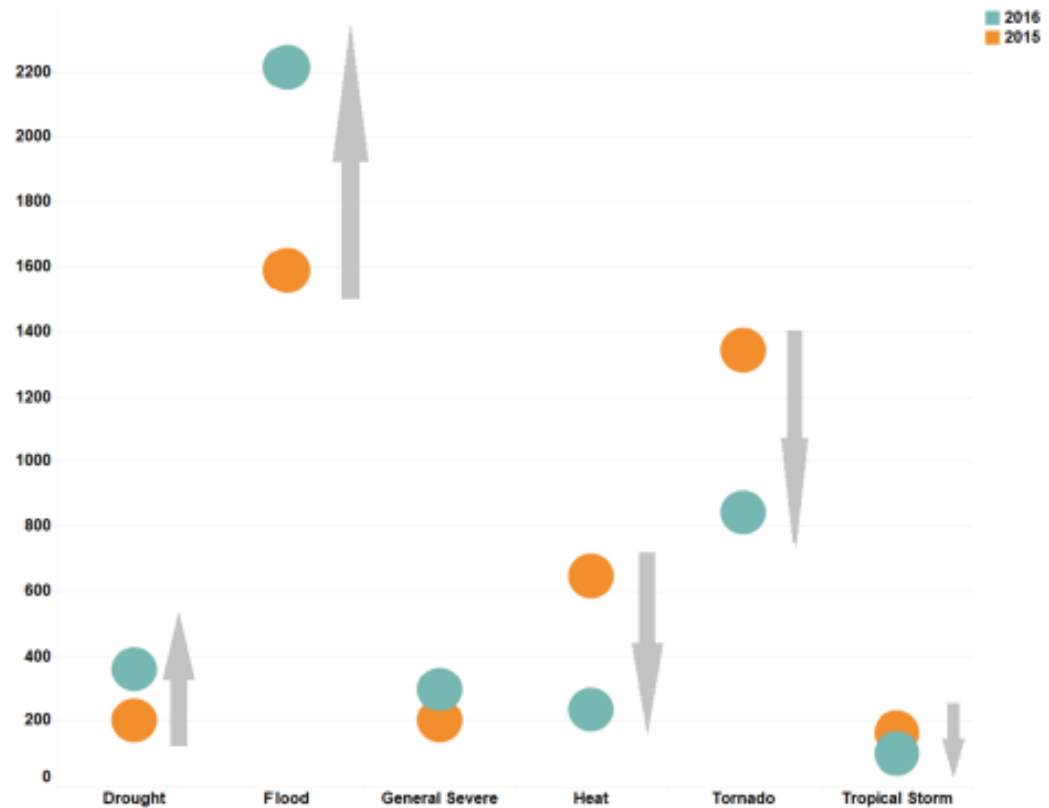
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IMPLEMENTATION – POWERPOINT PRESENTATION



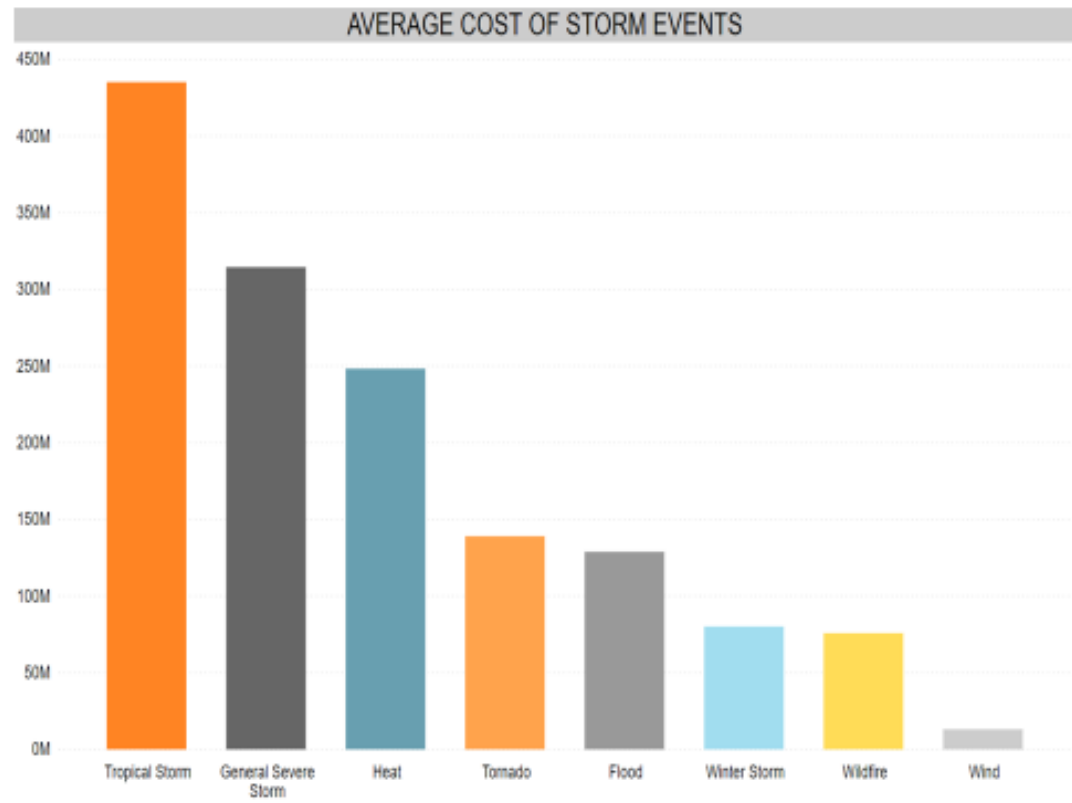
TEXAS WEATHER EVENTS IN RECENT YEARS

- In the recent years, Texas weather was plagued by drought, wildfires, and unprecedented floods (Texas Land Office, 2018).
- In 2010, drought and wildfires created conditions that reduced the ground's ability to absorb rain waters, which increased flood water accumulation during severe rainfalls (Texas Land Office, 2018). As drought conditions continued, wild fire occurrences increased. In 2011, there was over 30,000 wildfires in Texas (Kronzer, 2013). These events worsened land absorption conditions and increased flood water accumulation (Texas Land Office, 2018).
- Data analysis of top 5 costly storms show heat, tornado, and tropical storms to have a net decrease in storm count following the drought period. Flood occurrence increased significantly.



COSTS TO TEXAS

- Per NOAA (2018), a single storm event can potentially exceed \$2 billion in recovery costs.
- The average cost of recovery following a tropical storm can exceed \$400 million.
- Floods, tornados, and other severe storm events can cost Texas \$100 million to \$300 million in recovery costs.



DISASTER RECOVERY RESPONSIBILITY



- Primary responsibility is allocated to local emergency management teams within the Texas Department of Emergency Management (TDEM) (Kronzer, 2013).
- State and federal programs do not initiate disaster assistance until after local resources are exhausted (Kronzer, 2013).
- The agreement between the states and the Federal Emergency Management Agency (FEMA) allocates around 75% of federal assistance following a natural disaster when a need is determined (Kronzer, 2013).

RECOVER COST TO STORM EVENT COUNT BY REGION (2015-2016)

- All districts affected by storm events
- Houston area affected by the highest storm count, followed by Lubbock and Hurst/Garland.
- The average recovery cost for the top regions:
 - Houston - \$134 million
 - Lubbock - \$102 million
 - Hurst/Garland - 97 million

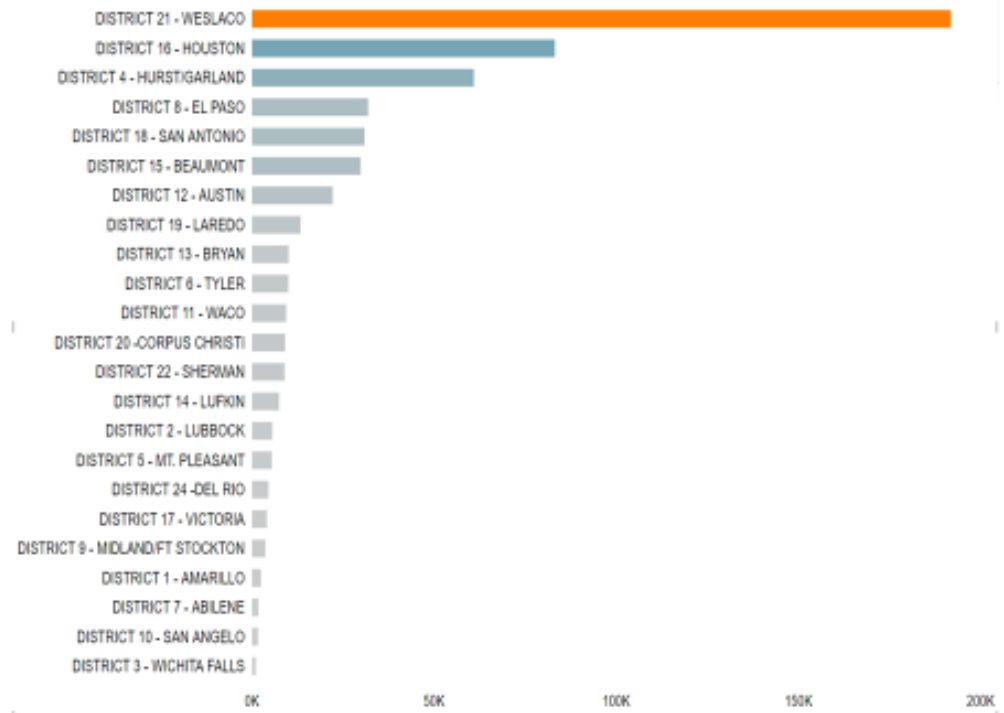
AVERAGE RECOVERY COST BY DISTRICT



DISASTER RECOVERY NEEDS – REGIONAL DEMOGRAPHICS

- Demographics are essential to recovery costs due to an increased need for financial assistance to low income disaster victims and/or renters (Texas Land Office, 2018).
- Poverty estimates are essential to budget allocation by region (Texas Land Office, 2018)
- The three regions with the high poverty estimates are:
 - District 21 – Weslaco
 - District 16 – Houston
 - District 4 – Hurst/Garland
- Houston and Hurst/Garland are the top two regions impacted by severe storms

POVERTY ESTIMATES BY DISTRICT



DISASTER RECOVERY SOCIAL SERVICE PROGRAMS

- Following a disaster the need for assistance is great. Consideration must be given to individuals homeless prior to an event, individuals displaced following an event, and individuals without means for repair (The Land Office, 2018). To meet these needs various social service programs are available.
- The costs of these programs exceed \$160 million (The Land Office, 2018).
- These programs are administered by the Texas Department of Housing and Community Affairs (TDHCA) (TDHCA, 2018).

SOCIAL SERVICE PROGRAMS	COSTS
Homeless Housing and Services Program and Continuum	\$15 million
Emergency Solutions Grants Program	\$17 million
Texas HOME Disaster Relief Program	\$10 million
Community Services Block Grant Program	\$120 million
TOTAL	\$162 million

DISASTER RECOVERY – ECONOMIC IMPACT

- The economic affects of a severe weather event are devastating.
 - Unemployment rates increase due to business closers (Texas Land Office, 2018)
 - Crop damage can delay the distribution of food and goods (Texas Land Office, 2018)
 - A reduction in tourism due to storm damage high tourist area (Texas Land Office, 2018)
- The total economic impact could cost Texas billions!





THANK YOU!

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- Feedback and Inquiries: Lakysa.linzy@snhu.edu

REFERENCES

- Kronzer, C. (2013). State disaster response and grant funding. Retrieved from http://www.lbb.state.tx.us/Documents/Publications/Issue_Briefs/298_State%20Disaster%20Funding.pdf
- NOAA (2018). Billion-dollar disasters: calculating the costs. Retrieved from <https://www.ncdc.noaa.gov/monitoring-references/dyk/billions-calculations>
- TDHCA, (2018). Help for Texas. Retrieved from <http://www.tdhca.state.tx.us/texans.htm>
- The Texas General Land Office, (2018). State of Texas plan for disaster recovery: hurricane harvey. Retrieved from <http://www.glo.texas.gov/the-glo/public-information/notices/files/action-plan-final-draft-1-18-18.pdf>

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IMPLEMENTATION – WEBPAGE



The Dangers of Flood Waters

[Home](#) [Flood Stats](#) [Response](#) [Preparation](#) [Flood Tips & Resources](#)



[\(1\)](#)

Death by Floods Compared to Other Storm Types

Death related storms in 2015 to 2016 were close to 200 in total^{[\(5\)](#)}. Flood waters were the top cause of deaths compared to a tornado and/or a tropical storm. Circumstances of flood related deaths are direct or indirect, and include drowning, heart attack, and risk-taking behavior.^{[\(4\)](#)}

STORM TYPE	%CT DEATHS
Flood	39.02%
Winter Storm	36.59%
General Severe Storm	9.76%
Tornado	7.32%
Heat	4.88%
Wind	2.44%
Total	100.00%

Circumstances for Deaths by Flood

The following are circumstances that could result in death because of flood conditions:⁽⁴⁾

- Losing control of vehicle while driving through flood waters
- Unable to rescue all vehicle passengers due to flood rush
- Loss of life while attempting to rescue victim from flood waters
- Being swept away by the current while wading/walking through flood waters
- Refusing to leave home and becoming trapped in attics in home by flood water
- Heart attacks caused by panic and stress
- Electrocution from downed power-lines

How to Respond in a Flood

At Home

If it's safe, stay home ⁽⁴⁾. It's better to be in a safe amount of water at home than out on the road where conditions will likely be worse ⁽⁴⁾. Utilize supplies from emergency kit and wait it out. Constantly listen to updates from media to ensure that if conditions in your area change you can respond promptly ⁽⁴⁾.

If home conditions are unsafe, leave and go to higher ground ⁽⁴⁾. **Do not go into the attic to escape flood waters unless you have a way to the roof from the attic!** ⁽⁴⁾. Many attics do not have a way out to the roof, and resident run the risk of becoming trapped in the attic with rising waters and no way-out ⁽⁴⁾. Do your best to get onto the roof or in a spot where you can be seen by rescue teams. ⁽⁴⁾

In vehicle

Avoid driving through flood waters if at all possible! If necessary to drive through flood waters, slowly navigate through the most shallow flow of water without stopping.⁽¹⁰⁾ A foot of water can float most vehicles, and two feet of rushing water can carry large vehicles away.⁽⁹⁾

If the vehicle is floating and you no longer have control, roll down window and call 911 if it's unsafe to exit the vehicle.⁽⁴⁾ **Do not exit a vehicle in rushing flood waters!**⁽⁴⁾ If unable to remain in vehicle, get to the roof of vehicle through an open window or sunroof, and remain on the roof until it is safe to go to ground. To avoid falling, stay low to the roof and do not stand⁽⁹⁾. Get to higher ground as soon as safe to do so.

Flood Types and How to Prepare

Flash Flood/Flood Warning

Warnings require ACTION! You should be moving to higher ground and completing necessary actions to get you and your loved ones to safety⁽⁶⁾. Flash floods happened quickly and often without time to prepare⁽⁸⁾. A flash floods can occur even if it's not raining⁽⁸⁾. Levee breaks and mudslides are examples of flash floods.⁽⁸⁾

Flood Watch

Watches require PREPARATION! You should be taking the time to prepare for flood warning conditions by coordinating plans with loved ones⁽⁷⁾. Things to do to prepare: ⁽³⁾

- Ensure emergency kits is fully stock for at least 72 hours (fill water containers, non-perishables, prescription medication, etc.)
- Move computers and other electronics to higher ground
- Move power strips from floors and unplug from power where contact with water is probable
- Move valuables to higher ground (pictures, etc.)

- Put passports and other important papers in plastic zip lock bags and place in freezer
- Turn off gas valves
- Fill bathtubs with clean water

Flood Advisory

Advisories are notifications of flood conditions that are unlikely to result in warnings [\(7\)](#). Take this time to prepare emergency kits, check flood insurance policy, install pumps, sandbagging outside doors, and cleaning gutters [\(2\)](#).

Flood Safety Tips & Resources

[Texas Department of Emergency Management's Flood Safety Awareness](#)

[Texas Department of Emergency Management's Flash Flood Safety Tips](#)

[Tips Emergency Home Kit](#)

[Tips Emergency Car Kit](#)

[Texas Agency Contacts](#)

Author: Lakysa Linzy

Affiliation: Southern New Hampshire University

Data Created: 04/15/2018

[Feedback Survey](#)

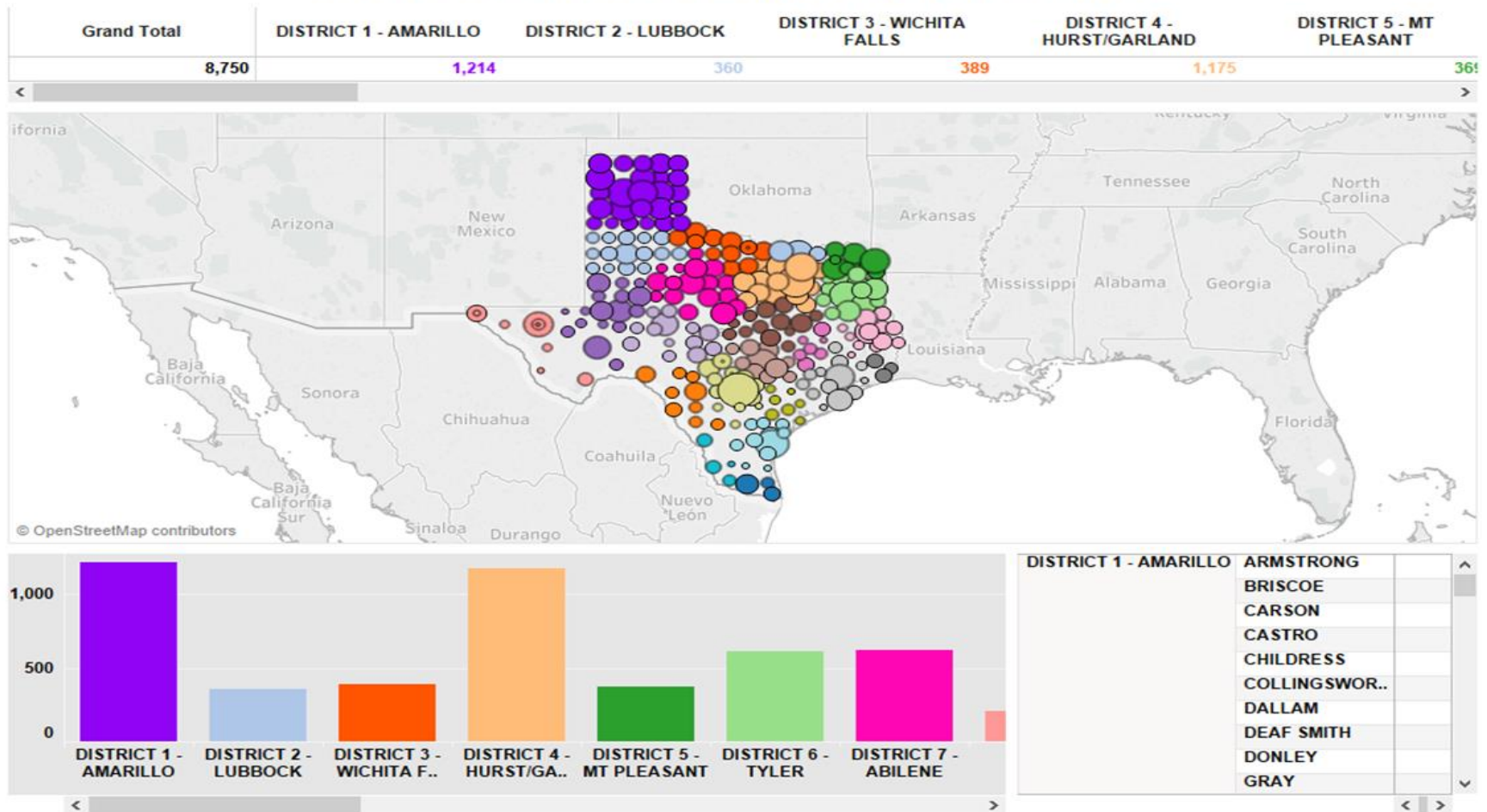
Bibliography

- (1) CBSDFW. (2017). People get stuck trying to drive through Houston flood waters. *Youtube*. [Source Link](#)
- (2) FEMA. (2014). How to prepare for a flood. [Source Link](#)
- (3) Government of Canada. (2017). Floods – what to do? [Source Link](#)
- (4) Gray, M. (2017). Stuck in the Texas floods? Here's what to do. *CNN*. [Source Link](#)
- (5) Jonkman, S & Kelman, I. (2005). An analysis of the causes and circumstances of flood disaster deaths. [Source Link](#)
- (6) NOAA. (2018). Storm events database. [Source Link](#)
- (7) US Department of Commerce, UDC. (2018). Flood warning vs. watch. *NOAA*. [Source Link](#)
- (8) US Department of Commerce, UDC. (2018). Flash flooding definition. *NOAA*. [Source Link](#)
- (9) Erdman, J. (2015). In flash flooding, your vehicle can be biggest danger. *The Weather Channel*. [Source Link](#)
- (10) Telegraph Reporters (2018). How to drive through standing water and floods (if you really have to). [Source Link](#)

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IMPLEMENTATION – INTERACTIVE DASHBOARD ONE

EVENT COUNT BY TDEM DISTRICT/COUNTY



Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

DESCRIPTION

Map based on longitude (generated) and latitude (generated). Color shows details about district. Size shows count of storm type. Details are shown for district and county. The data is filtered on action (county, district) and action (district). The action (county, district) filter keeps 269 members. The action (district) filter keeps 23 members.

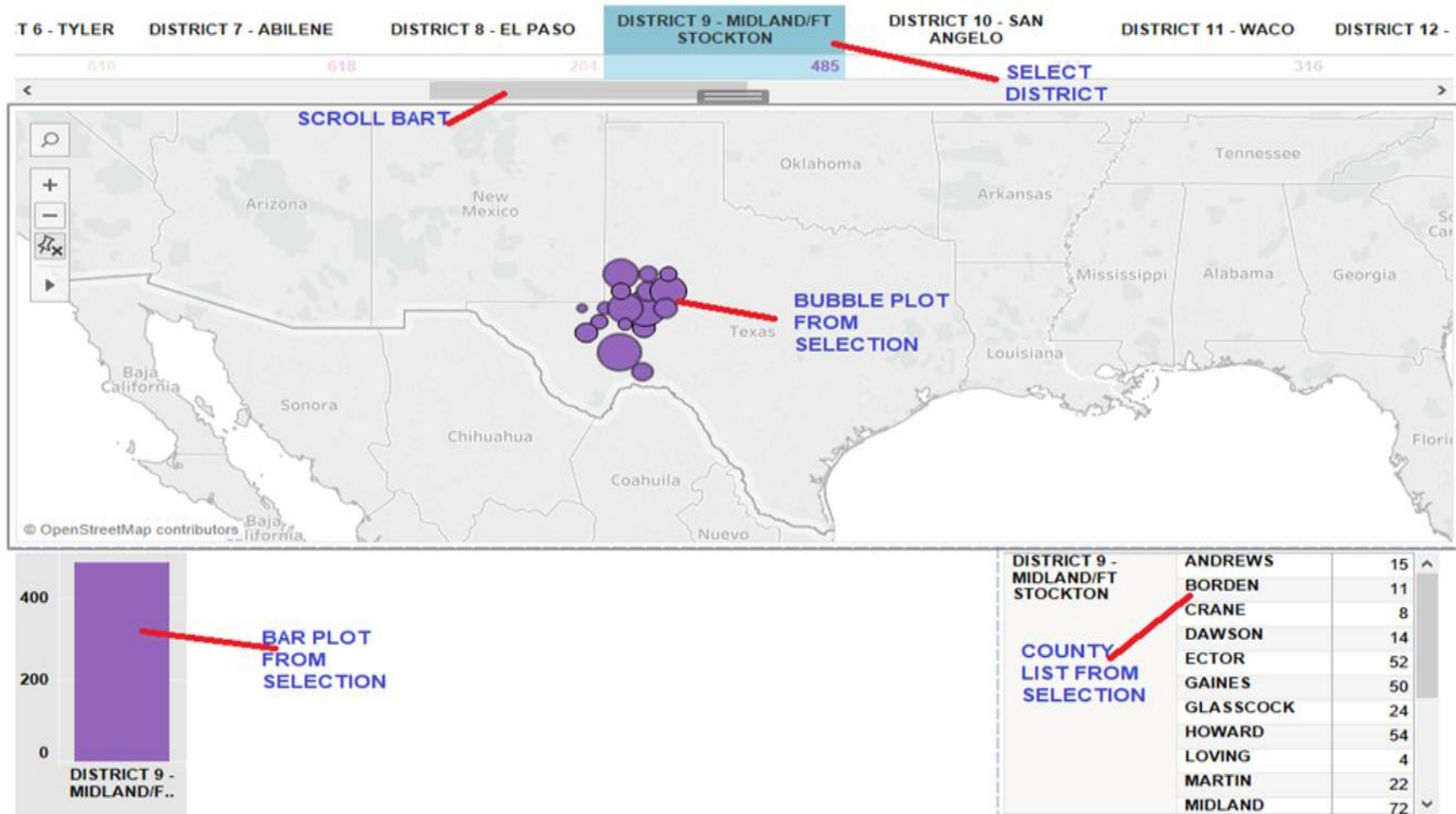
USAGE/INTERACTIONS

The first chart is a horizontal legend with total storm event counts by TDEM district. Each district listed in the legend shows the total storm event count for that district. The second chart is a map chart colored by TDEM districts and sized by event count per county.

The third chart is a bar chart colored and scaled to correspond with the legend map selection. The forth chart is a static chart detailing the event count by county. All charts on this dashboard connect and respond to selections as one. To interact with dashboard:

- Use scroll bar on the legend to navigate to the desired district
- Click to select desired district
- Look below to view the selected district by region on the map chart, by quantity on the bar chart, and by county breakdown using static chart
- Zoom in on selected area to view the county name for each bubble to compare bubble size. Hover over the bubble to view tooltip listing county, district, and storm event count.

EVENT COUNT BY TDEM DISTRICT/COUNTY



Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

- Hover over the bar for tooltip showing district and storm event count for that selection.
- If no legend item is selected, click bar for backward drill down the for the selection.

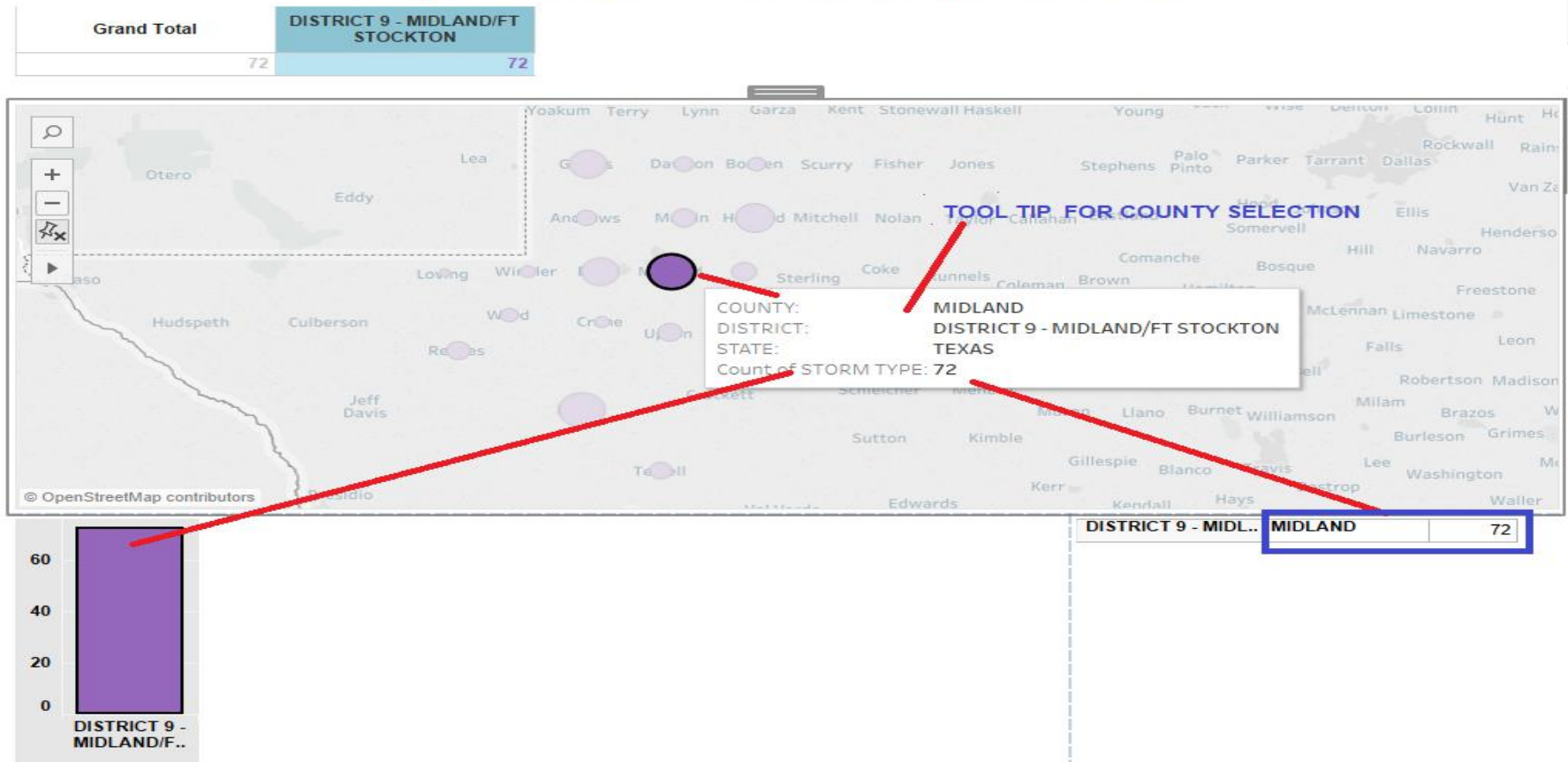
EVENT COUNT BY TDEM DISTRICT/COUNTY



Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

- Click desired county to drill down for that county on all four charts.

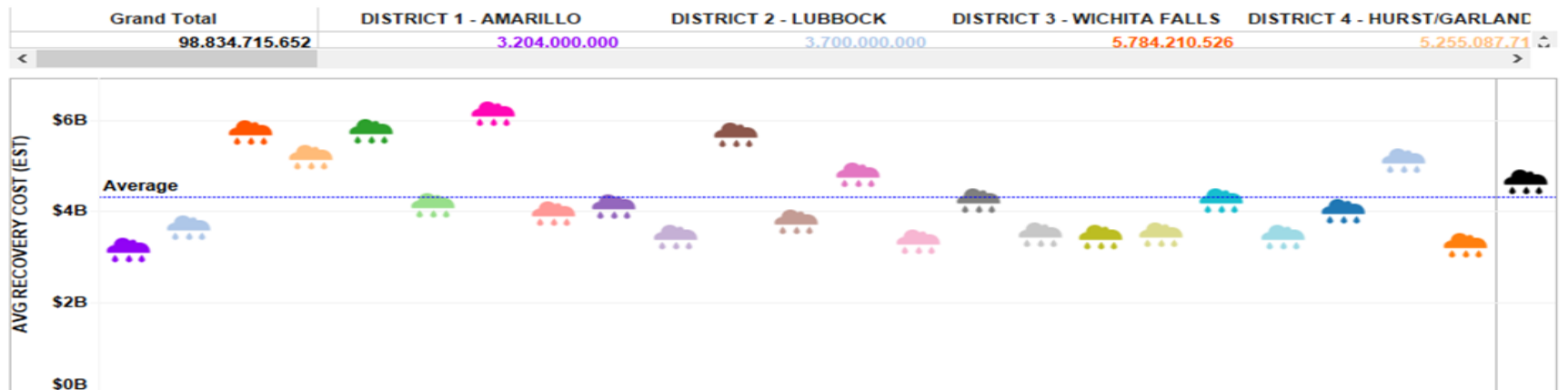
EVENT COUNT BY TDEM DISTRICT/COUNTY



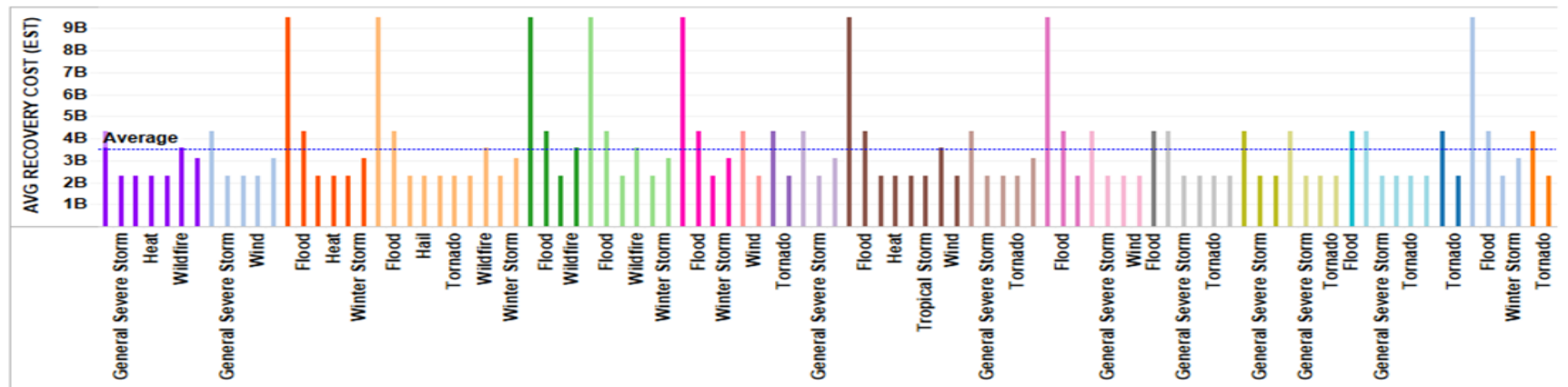
Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

IMPLEMENTATION – INTERACTIVE DASHBOARD TWO

AVG RECOVERY COST BY TDEM DISTRICT



AVG COST BY STORM TYPE



Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

DESCRIPTION

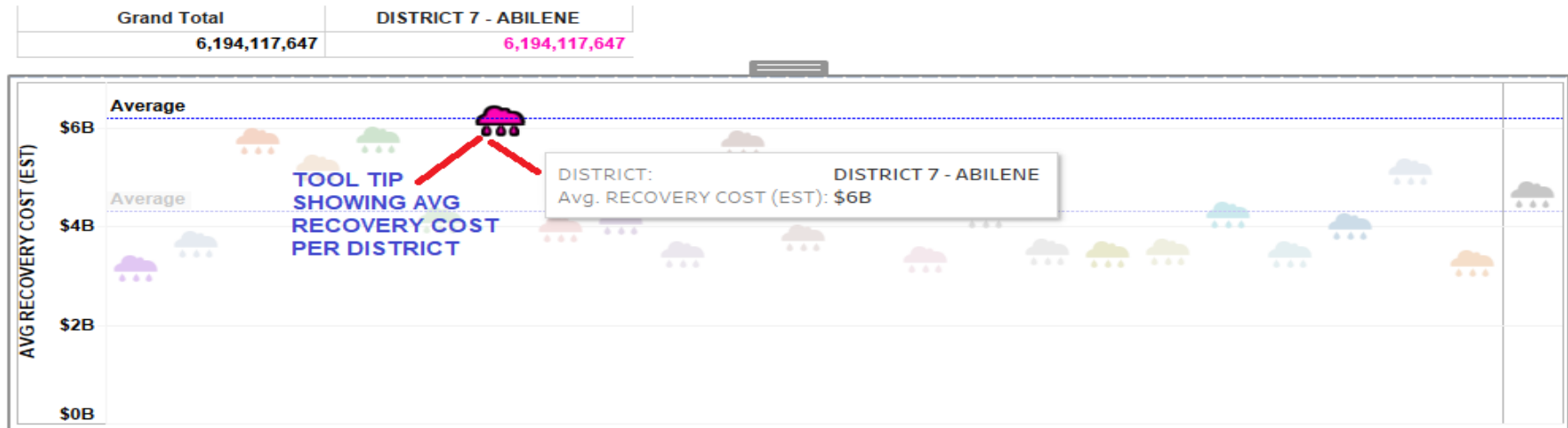
Average of recovery cost (est.) for each district. Color shows details about district. The data is filtered on action (district) and action (district, storm type). The action (district) filter keeps 23 members. The action (district, storm type) filter keeps 187 members. All four charts connect and change with each interaction.

USAGE/INTERACTIONS

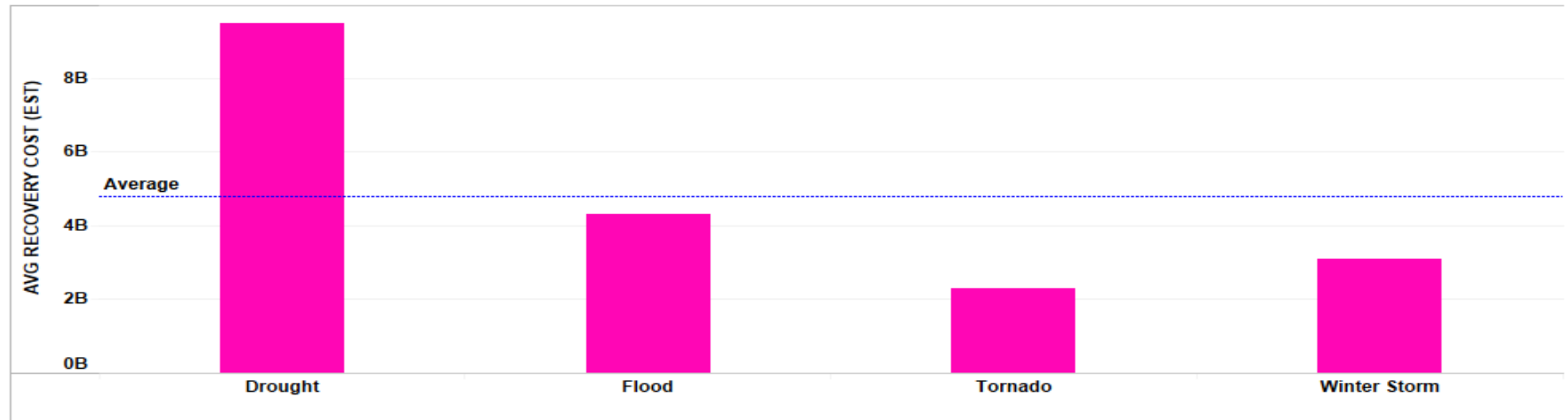
The first chart is a horizontal legend with total recovery costs by TDEM district. Each district listed in the legend shows the total recovery costs for that district. The second chart is a line chart colored by TDEM districts and distributed by recovery costs per county. The third chart is a bar graph demonstrating recovery costs by storm event type. All charts on this dashboard connect and respond to selections as one. To interact with dashboard:

- Use scroll bar on the legend to navigate to the desired district
- Click to select desired district
- Look below to view the selected quantities for the average recovery costs on the line chart and bar chart to see the comparison in cost between districts and the drill down of average costs per storm type within the selected district
- Hover over any color item for a tooltip detailing district and average cost per district.

AVG RECOVERY COST BY TDEM DISTRICT

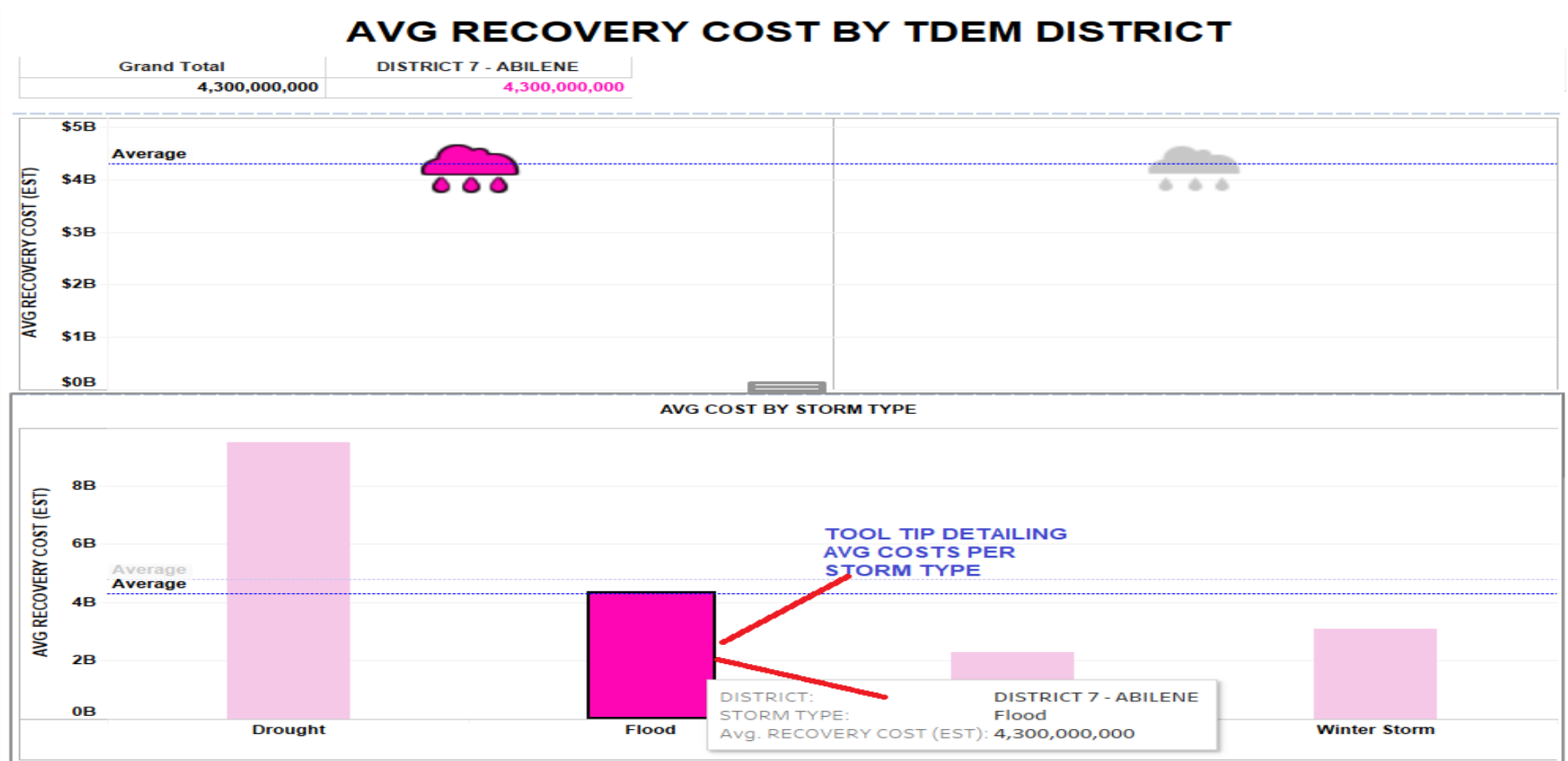


AVG COST BY STORM TYPE



Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

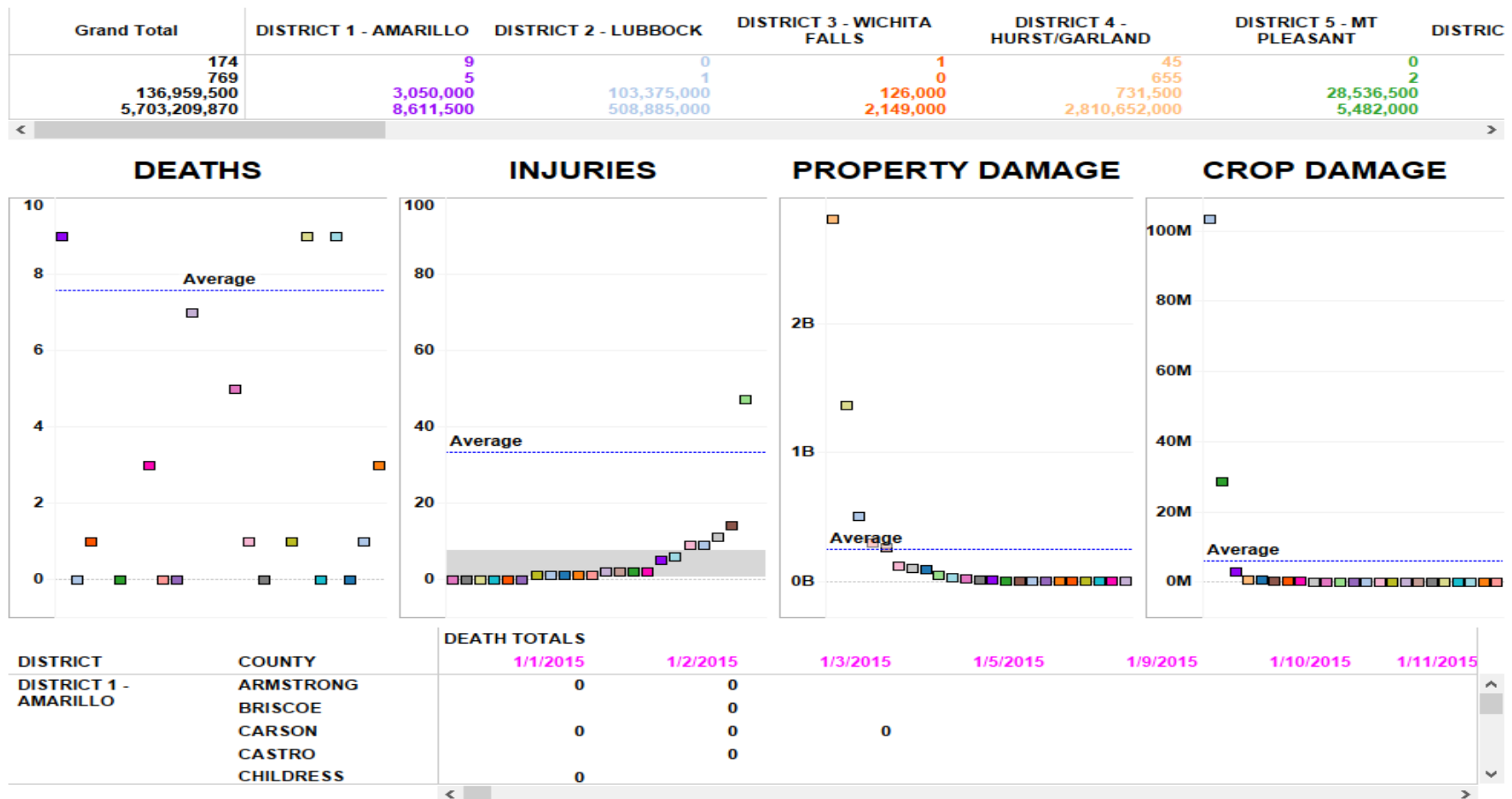
- To drill down further click on a bar representing the desired storm type to view the average recovery costs for that storm type.
- Hover over bar to get tool tip detailing district, storm type, and average recovery costs



Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

IMPLEMENTATION – INTERACTIVE DASHBOARD THREE

AFFECTS OF STORM BY TDEM DISTRICT



Author: Lakysa Linzy, Southern New Hampshire University, created on 04/08/2018. For questions/ problems with this dashboard, contact: 1-800-123-1234. Feedback Survey: <https://www.surveymonkey.com/r/3B76Z5L>. NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>.

DESCRIPTION

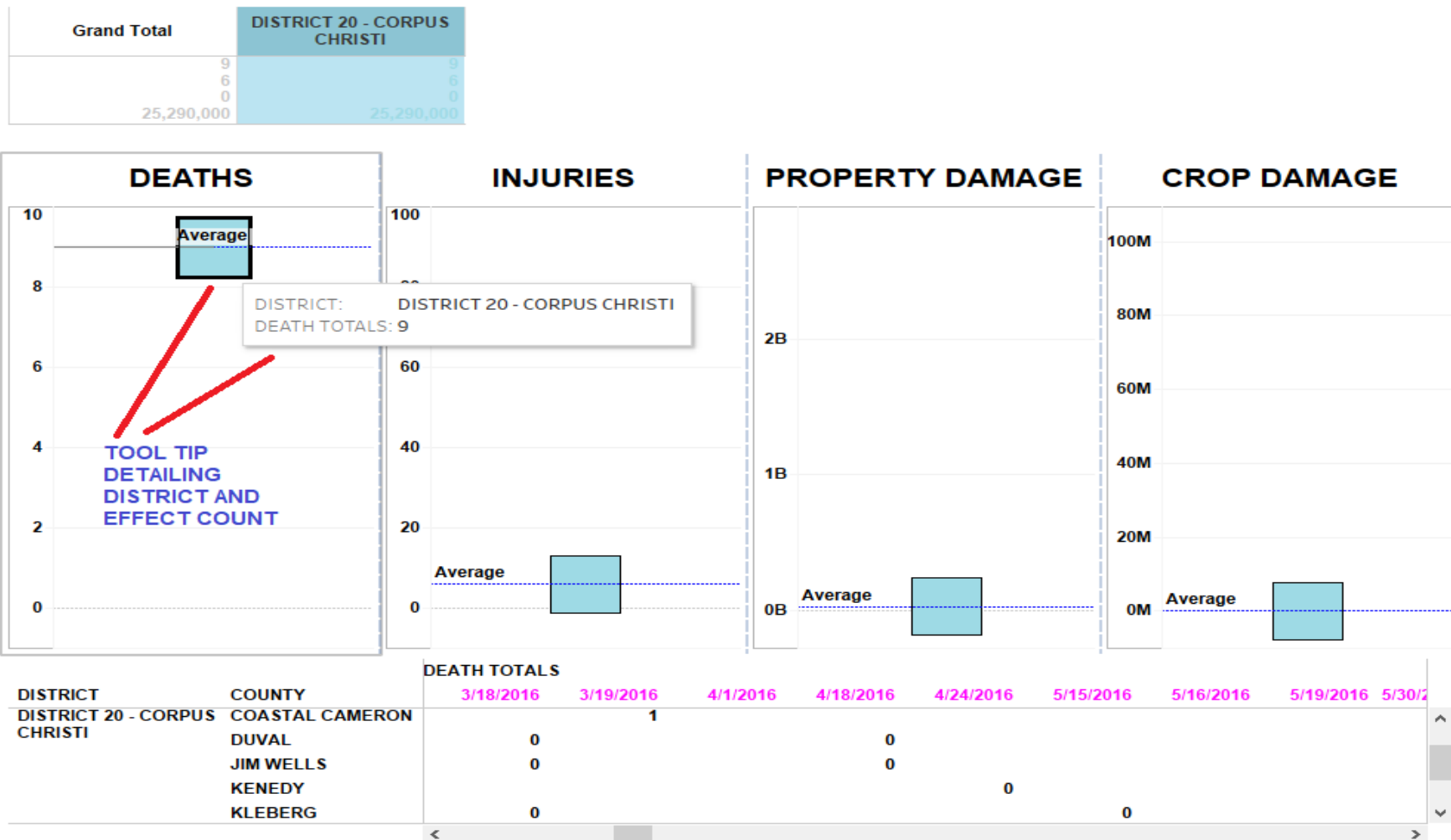
Death totals, injury totals, property damage totals and crop damage totals broken down by date of event vs. District and county. The data is filtered on narrative of event and date of event day. The narrative of event filter excludes null. The date of event day filter has multiple members selected.

USAGE/INTERACTIONS

The first chart is a horizontal legend with total death totals, injury totals, property damage totals and crop damage by TDEM district. The second through forth charts are scatter plots colored by TDEM districts. The scatter plots demonstrate deaths, injuries, property damage, and crop damage counts per district. The fifth chart is a static chart detailing death totals, injury totals, property damage totals, and crop damage totals by county. The fifth chart further drills down by date of event and narrative of event conditions. All charts on this dashboard connect and respond to selections as one. To interact with dashboard:

- Use scroll bar on the legend to navigate to the desired district
- Click to select desired district
- Look below to view the selected counts for death totals, injury totals, property damage totals, and crop damage totals to see the comparison in cost between districts and the drill down of average costs per event effect/damage type
- Hover over any color item for a tooltip detailing district and event effects/damage per district.

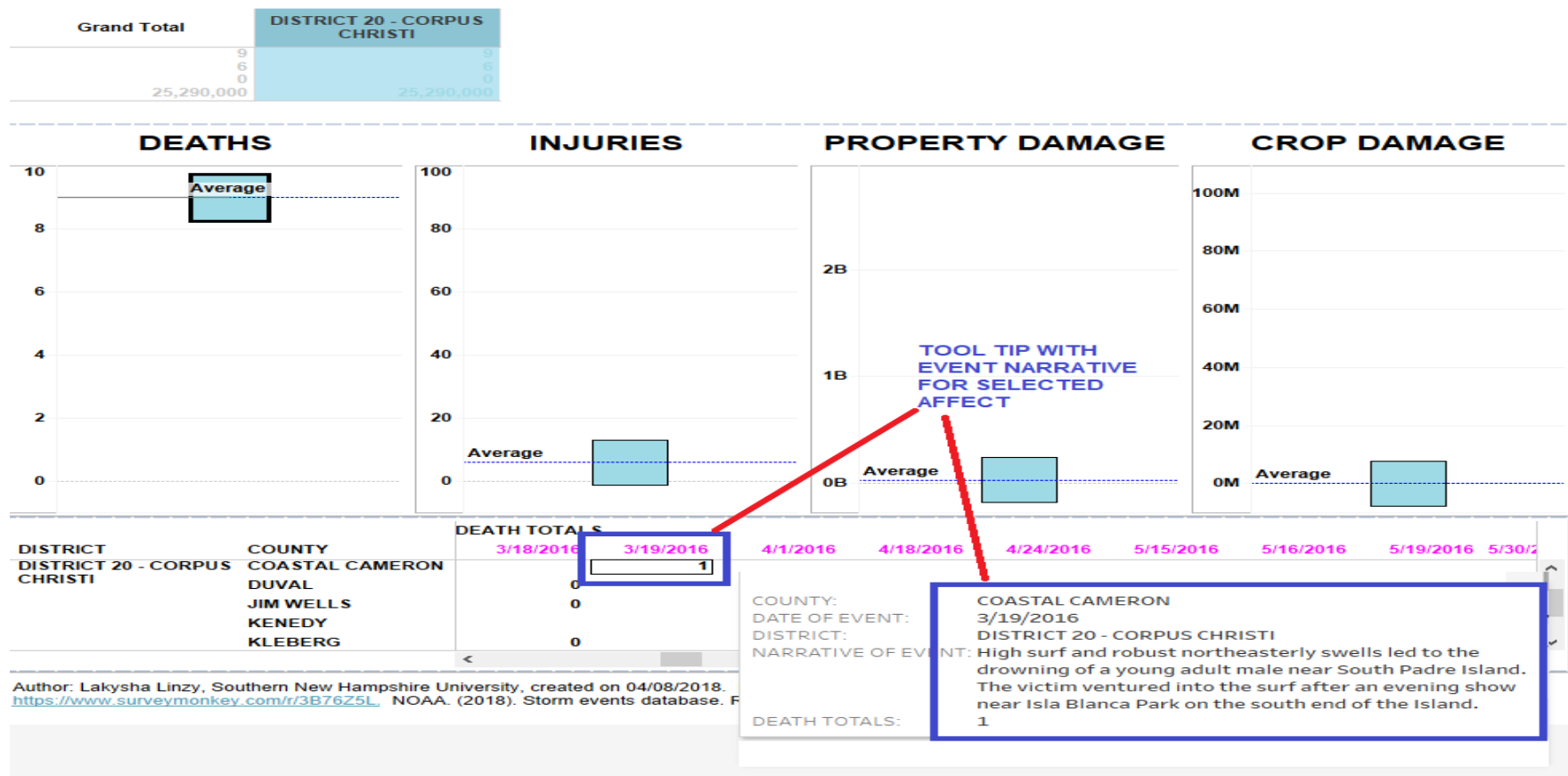
AFFECTS OF STORM BY TDEM DISTRICT



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- To drill down further use fifth chart (static chart) to scroll over to desired effect type and date to view effect counts by county
- Hover over effect count to view tooltip containing date of event and its narrative of conditions

AFFECTS OF STORM BY TDEM DISTRICT



- For additional information or assistance with the dashboards, view contact information printed at the bottom of dashboard
- Please provide feedback on you experience with the dashboards by clicking the link to the online survey printed at the bottom of each dashboard.

Thank you for viewing!

REFERENCES

- Hearn, P. (2017). Powerpoint presentations vs. online dashboards – which is better? *MRDC Software*. Retrieved from <http://www.mrdcsoftware.com/blog/powerpoint-presentations-vs-online-dashboards>
- MacAloney, B. (2016). National weather service instruction 10-1605. Retrieved from <http://www.nws.noaa.gov/directives/sym/pd01016005curr.pdf>
- NCEI (2018). Storm data faq page. NOAA. Retrieved from <https://www.ncdc.noaa.gov/stormevents/faq.jsp>
- NOAA. (2018). Storm events database. Retrieved from <https://www.ncdc.noaa.gov/stormevents>
- TDEM (2018). TDEM mission, organization & responsibilities. *Texas Department of Public Safety*. Retrieved from <https://www.dps.texas.gov/dem/mission.htm>
- Texas Impact. (2015). The Texas budget process. *Texas Interfaith Center for Public Policy*. Retrieved from <https://texasimpact.org/content/texas-budget-process>